LAB NO:04

Computer Mouse: Definition, Types, and Circuit Diagram

1. Definition of a Computer Mouse

A computer mouse is an input device used to control the movement of the cursor or pointer on a computer screen. It allows the user to select items, open files, click buttons, drag objects, and perform other operations.

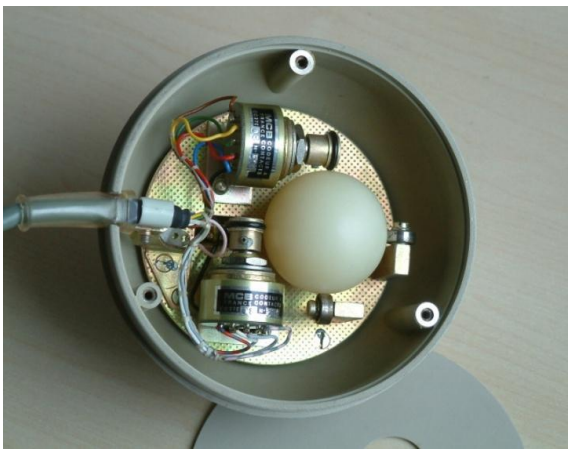
A mouse is commonly connected to a computer through a USB cable or wirelessly using Bluetooth or a USB receiver.







2. Types of Computer Mouse



1. Mechanical Mouse

A mechanical mouse uses a rubber or metal ball underneath to detect movement. It is an older type of mouse.



2. Optical Mouse

An optical mouse uses an LED light and an optical sensor to detect movement. It is commonly used with computers today.



3. Laser Mouse

A laser mouse uses a laser-based sensor to detect movement and can work on many different surfaces.



4. Wired Mouse

A wired mouse connects directly to a computer using a cable, usually a USB cable.



5. Wireless Mouse

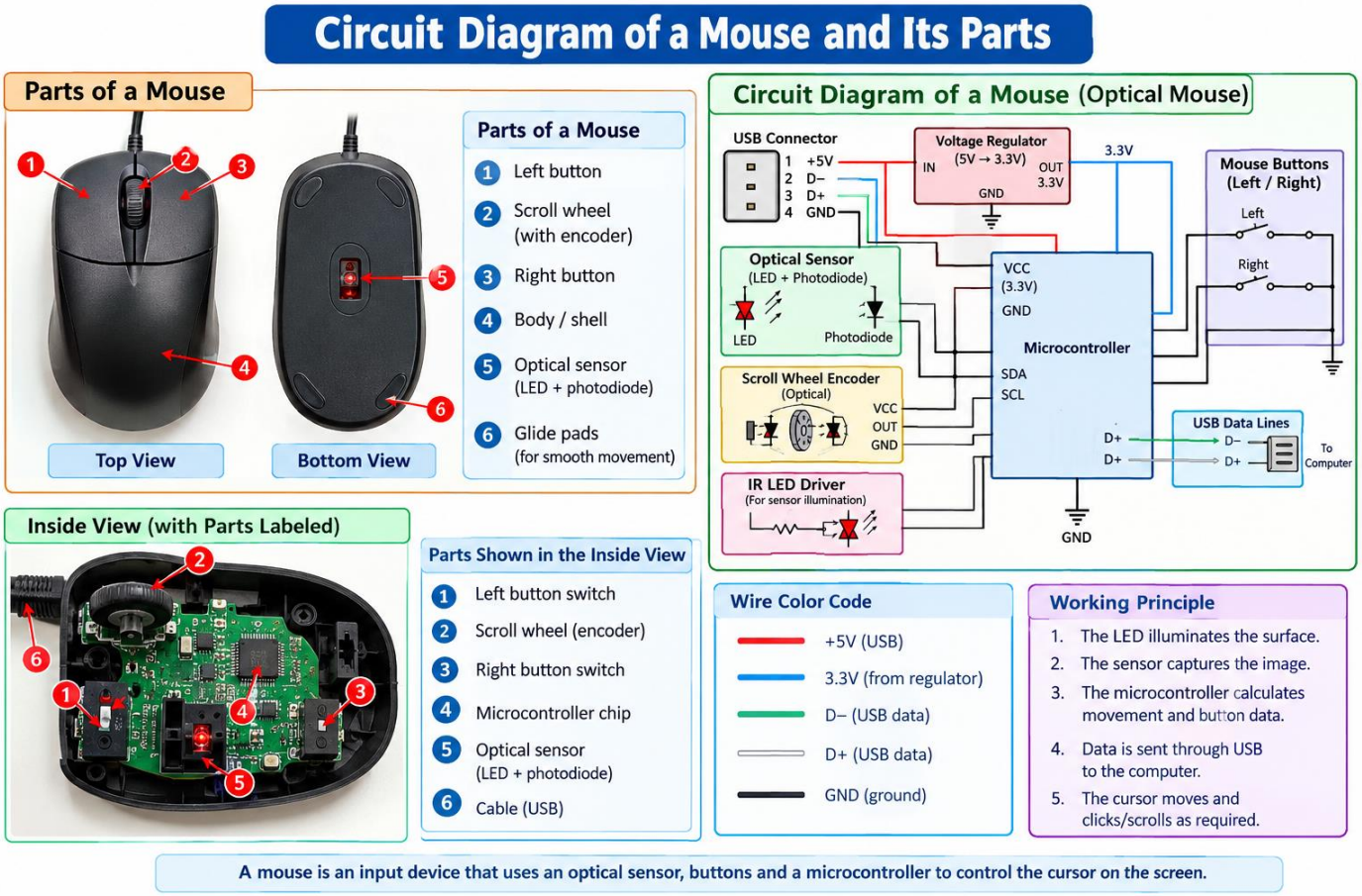
A wireless mouse communicates with a computer using Bluetooth or a wireless USB receiver.



6. Ergonomic Mouse

An ergonomic mouse is designed to provide a more comfortable hand position and may help reduce wrist strain.

3. Circuit Diagram of an Optical Mouse



Main components of the mouse circuit

1. USB connector: Supplies power and transfers data between the mouse and computer.
2. Optical sensor: Detects movement across a surface.
3. Microcontroller: Processes movement and button signals.
4. Push-button switches: Detect left-click, right-click, and sometimes middle-click.
5. Scroll-wheel encoder: Detects scrolling in mice that have a wheel.
6. LED or laser: Illuminates the surface so the sensor can track movement.
7. PCB (Printed Circuit Board): Holds and connects the electronic components.

4. Working Principle of a Mouse

1. The user moves the mouse across a surface.
2. The optical sensor detects changes in the surface image.
3. The controller calculates the direction and distance of movement.
4. The controller sends movement and button data to the computer.

5. The cursor moves on the screen, and clicking or scrolling performs the required action.

5. Conclusion

A computer mouse is an important input device used to control a computer. The main types include mechanical, optical, laser, wired, wireless, and ergonomic mice. Its electronic circuit detects movement and button presses and sends the resulting information to the computer.